

Load Interconnection Request System Impact Study Report

For Queue Project Q11-0084

Mccabe 138 kV
November 15, 2012

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q11-0084. The Interconnection Customer proposes a 390.2 MW large load (data center) facility interconnecting at the McCabe 138 kV substation in the North Carolina area. The request was received on November 15, 2011. The study evaluated system intact and single contingency conditions using the regional planning case, including 74 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$270,526,000.

3. Network Impacts

3.1 Load Deliverability Analysis

The Phelon Parts 138/69 kV Autotransformer is overloaded to 252.7% of its 60 MVA emergency rating for the loss of the Home 138/1 kV XFMR. The project contributes 47.9% DFAX (186.8 MW) to this overload.

The McCabe – Laurel City 138 kV is overloaded to 212.6% of its 191 MVA emergency rating for the loss of the McCabe Substation 138/1 kV TX. The project contributes 100.0% DFAX (390.2 MW) to this overload.

Mccabe 138-1 kV XFMR: 184.6% loading (220 MVA) following the outage of the Laurel City – McCabe 138 kV L; DFAX = 100.0%.

For the loss of Mira Vista – Yarnen 345 kV L, the Elmo – Robinson Avenue 138 kV line loads to 150.8% of its 250 MVA rating. Project DFAX: 29.7%.

The No Mans Land – Ne-Hobbs 138 kV L is overloaded to 147.2% of its 261 MVA emergency rating for the loss of the Hartselle Substation 345/138kV Transformer. The project contributes 62.0% DFAX (241.8 MW) to this overload.

The Summit Farms Solar – Robinson Avenue Substation 138 kV is overloaded to 145.5% of its 284 MVA emergency rating for the loss of the Yarnen – Mira Vista 345 kV Line. The project contributes 29.7% DFAX (116.0 MW) to this overload.

Ne-Hobbs – Harbor Beach Substation 138 kV: 143.9% loading (248 MVA) following the outage of the Hartselle 345/138 kV XFMR; DFAX = 62.0%.

The Hartselle – Summit Farms Solar 138 kV is overloaded to 142.2% of its 297 MVA emergency rating for the loss of the Mira Vista – Yarnen 345 kV. The project contributes 29.7% DFAX (115.9 MW) to this overload.

For the loss of McCartney – Yarnen 345 kV, the Bobville Substation – Avra Vly 138 kV loads to 132.2% of its 235 MVA rating. Project DFAX: 11.0%.

The Harbor Beach – Hickory 138 kV Line is overloaded to 131.3% of its 238 MVA emergency rating for the loss of the Hartselle Substation 345/138 kV Autotransformer. The project contributes 62.0% DFAX (241.8 MW) to this overload.

The Hardy eit – Battlefield 138 kV Line is overloaded to 131.1% of its 250 MVA emergency rating for the loss of the Yarnen – Mia Vista 345 kV. The project contributes 9.7% DFAX (37.9 MW) to this overload.

Synata Hugoton – Battlefield 138 kV: 128.5% loading (230 MVA) following the outage of the Mira Vista – Yarnen 345 kV; DFAX = 9.7%.

For the loss of Mira Vista – Yarnen 345 kV, the Crittenden – Kraft 138 kV loads to 127.8% of its 408 MVA rating. Project DFAX: 10.9%.

The Roanoke Rapids 345 kV/138 kV XFMR is overloaded to 124.0% of its 676 MVA emergency rating for the loss of the Mccartney – Yarnen 345 kV. The project contributes 26.2% DFAX (102.2 MW) to this overload.

The Vereda 3103 – Del Rosa 138 kV is overloaded to 122.7% of its 483 MVA emergency rating for the loss of the Mickens – Mira Vista 345 kV. The project contributes 15.4% DFAX (60.1 MW) to this overload.

For the loss of Yarnen – Mira Vista 345 kV, the Chuckatuck – Kraft 138 kV L loads to 122.6% of its 461 MVA rating. Project DFAX: 10.9%.

Synata Hugoton – Lightfoot 138 kV Line: 121.5% loading (220 MVA) following the outage of the Yarnen – Mira Vista 345 kV; DFAX = 9.7%.

The Portland FI – Bonny Brook 138 kV is overloaded to 121.0% of its 195 MVA emergency rating for the loss of the Mira Vista – Yarnen 345 kV. The project contributes 6.6% DFAX (25.7 MW) to this overload.

Tillman – Carroll 138 kV: 119.5% loading (284 MVA) following the outage of the Yarnen Substation – Mira Vista 345 kV; DFAX = 8.7%.

For the loss of Mira Vista – Yarnen 345 kV L, the Hardy Reit – Stepdown 138 kV loads to 118.8% of its 299 MVA rating. Project DFAX: 5.6%.

Mira Vista 138 kV/1 kV TX: 118.3% loading (524 MVA) following the outage of the Mickens – Mira Vista 345 kV; DFAX = 15.4%.

Del Rosa – Mira Vista 138 kV: 117.9% loading (525 MVA) following the outage of the Mickens – Mira Vista 345 kV; DFAX = 15.4%.

The Danc Lfgte Facility – Bonny Brook 138 kV is overloaded to 117.5% of its 209 MVA emergency rating for the loss of the Yarnen – Mira Vista 345 kV. The project contributes 6.6% DFAX (25.7 MW) to this overload.

The Bobville – Cochise Substation 138 kV is overloaded to 116.2% of its 237 MVA emergency rating for the loss of the Mccartney – Yarnen 345 kV. The project contributes 5.9% DFAX (23.2 MW) to this overload.

The Roanoke Rapids Substation – Fyffe 138 kV #2 is overloaded to 116.1% of its 413 MVA emergency rating for the loss of the Mccartney – Yarnen 345 kV L. The project contributes 13.9% DFAX (54.4 MW) to this overload.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Phelon Parts 138/69 kV Autotransformer	Replace transformer with higher-rated unit	\$4,939,000	100% (\$4,939,000)
Mccabe – Laurel City 138 kV	Rebuild line with higher-capacity conductor	\$52,846,000	100% (\$52,846,000)
Mccabe 138/1kV Autotransformer	Replace transformer with higher-rated unit	\$4,097,000	100% (\$4,097,000)
Elmo – Robinson Avenue 138 kV line	Rebuild line with higher-capacity conductor	\$4,352,000	91% (\$3,952,000)
No Mans Land – Ne-Hobbs 138 kV	Rebuild line with higher-capacity conductor	\$5,679,000	100% (\$5,679,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Summit Farms Solar – Robinson Avenue Substation 138 kV	Rebuild line with higher-capacity conductor	\$8,468,000	91% (\$7,690,000)
Ne-Hobbs – Harbor Beach 138 kV	Rebuild line with higher-capacity conductor	\$6,209,000	100% (\$6,209,000)
Hartselle – Summit Farms Solar 138 kV	Rebuild line with higher-capacity conductor	\$5,710,000	91% (\$5,186,000)
Bobville – Avra Valley 138 kV	Rebuild line with higher-capacity conductor	\$12,166,000	78% (\$9,540,000)
Harbor Beach – Hickory 138 kV Line	Rebuild line with higher-capacity conductor	\$9,265,000	100% (\$9,265,000)
Hardy eit – Battlefield 138 kV Line	Rebuild line with higher-capacity conductor	\$27,202,000	74% (\$20,062,000)
Synata Hugoton – Battlefield 138 kV	Rebuild line with higher-capacity conductor	\$1,777,000	74% (\$1,311,000)
Crittenden – Kraft 138 kV	Rebuild line with higher-capacity conductor	\$7,808,000	68% (\$5,342,000)
Roanoke Rapids 345-138 kV Transformer Bank	Replace transformer with higher-rated unit	\$19,528,000	81% (\$15,856,000)
Vereda 3103 – Del Rosa 138 kV	Rebuild line with higher-capacity conductor	\$2,203,000	74% (\$1,619,000)
Chuckatuck – Kraft 138 kV L	Rebuild line with higher-capacity conductor	\$8,792,000	68% (\$6,015,000)
Lightfoot Substation – Synata Hugoton 138 kV	Rebuild line with higher-capacity conductor	\$5,950,000	74% (\$4,388,000)
Portland FI – Bonny Book 138 kV	Rebuild line with higher-capacity conductor	\$46,732,000	74% (\$34,590,000)
Tillman – Carroll 138 kV	Rebuild line with higher-capacity conductor	\$5,708,000	68% (\$3,893,000)
Hardy Reit – Stepdown 138 kV	Rebuild line with higher-capacity conductor	\$6,456,000	72% (\$4,627,000)
Mira Vista 138/1 kV Transformer Bank	Replace transformer with higher-rated unit	\$4,872,000	74% (\$3,581,000)
Del Rosa – Mira Vista 138 kV	Rebuild line with higher-capacity conductor	\$3,849,000	74% (\$2,829,000)
Bonny Brook – Danc Lfge Facility 138 kV	Rebuild line with higher-capacity conductor	\$14,247,000	74% (\$10,545,000)
Bobville – Cochise Substation 138 kV	Rebuild line with higher-capacity conductor	\$16,755,000	77% (\$12,933,000)
Roanoke Rapids Substation – Fyffe 138 kV #2	Rebuild line with higher-capacity conductor	\$41,464,000	81% (\$33,532,000)
	Total allocated to project		\$270,526,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.